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Case Study: Bellabeat

You have been asked to focus on one of Bellabeat's products and analyze smart device data to gain insight into how consumers are using their smart devices. The insights you discover will then help guide marketing strategy for the company. You will present your analysis to the Bellabeat executive team along with your high-level recommendations for Bellabeat's marketing strategy.

You joined this team six months ago and have been busy learning about Bellabeat's mission and business goals — as well as how you, as a junior data analyst, can help Bellabeat achieve them.

Note: Completing this case study within a week is a good goal.

Ask

These questions will guide your analysis:

1. What are some trends in smart device usage?
 2. How could these trends apply to Bellabeat customers?
 3. How could these trends help influence Bellabeat marketing strategy?
- What do these trends tell us about how the customers use their Bellabeat smart device?
 - What percentage of people used the app for less than an hour a day?
 - What metrics tell us that the customer is sedentary, active, or very active?
 - Are there any outliers in trends in the past 5 years?

You will produce a report with the following deliverables:

1. A clear summary of the business task
 - Analyze one of Bellabeat's smart device's data to gain insight into how consumers are using their smart device, the Bellabeat leaf. Then use the insights you discover to help guide marketing strategy for the company, and provide marketing strategy recommendations to the executive team.
2. A description of all data sources used
3. Documentation of any cleaning or manipulation of data
4. A summary of your analysis

5. Supporting visualizations and key findings

6. Your top high-level content recommendations based on your analysis

Case Study Roadmap –

Ask Guiding questions

- What is the problem you are trying to solve?

- I want to find out how consumers are using one of Bellabeat's smart devices in order to guide the company's marketing strategy.

- How can your insights drive business decisions?

- Trends can tell us how to improve app functionality or advertise to a specific audience

Key tasks

- Identify the business task

- Analyze one of Bellabeat's smart device's data to gain insight into how consumers are using their smart device. Then use the insights you discover to help guide marketing strategy for the company, and provide marketing strategy recommendations to the executive team.

- Consider key stakeholders

- The stakeholders are Urska Srsen (cofounder and CCO), Sando Mur (cofounder and key member of the executive team), and Bellabeat's analytics team

Deliverable

- A clear statement of the business task

- Analyze one of Bellabeat's smart device's data to gain insight into how consumers are using their smart device. Then use the insights you discover to help guide marketing strategy for the company, and provide marketing strategy recommendations to the executive team. Simply, it would be: How do Bellabeat's users use Bellabeat smart devices?

Prepare

[FitBit Fitness Tracker Data](#) (CC0: Public Domain, data set made available through Mobius):

This Kaggle data set contains personal fitness tracker data from thirty fitbit users. Thirty eligible Fitbit users consented to the submission of personal tracker data, including minute-level output for physical activity, heart rate, and sleep monitoring. It includes information about daily activity, steps, and heart rate that can be used to explore users' habits.

Sršen tells you that this data set might have some limitations and encourages you to consider adding another data set to help address those limitations as you begin to work more with this data.

Case Study Roadmap -

Prepare Guiding questions

● Where is your data located?

-Kaggle: FitBit Fitness Tracker Data. It is stored locally on my computer now.

● How is the data organized?

-The data is organized in multiple tables split by files. Each file captures a different aspect of Fitbit user behavior. Every table has an ID and date column. The data is considered long since each row is one time point per subject, so each subject will have data in multiple rows. There are two datasets: one for 3-12-16, and one for 4-12-16.

● Are there issues with bias or credibility in this data? Does your data ROCCC?

-The data comes from the measurements of the app itself. Respondents of the survey only consented to give their information so there was no self-reporting. But there could be some bias because there is variation between output apparent in the use of different fitbit trackers and individual tracking preferences. Since there were only 30 respondents, it is possible there could be sampling bias.

Reliable: For the purposes of this case study, yes. This data has been vetted by Google, being a part of the course, and Kaggle, with a Usability score of 9.41. It has a CC0: Public Domain license. For purposes of making actual business decisions, it is limited because there are only about 30 users.

Original: No. This is a third-party dataset because it comes from Amazon Mechanical Turk.

Comprehensive: Yes, mostly. There is information on the user's steps, sleep, heartrate, and calories burned, with dates and IDs to match information across table. However, there are columns that categorize users as sedentary, active, or very active, which isn't clear whether it is self-reported in the app or if the app itself puts a user in a category based on their usage. Also, the dataset does not include gender, age, income, location, or Bellabeat customer data, so recommendations should be high-level.

Current: It depends. For the time frame, in 2016, this information is accurate, so analysis will be under the assumption that the business task was asked for the year 2016, otherwise this dataset is not current.

Cited: Yes. Mobius, a Kaggle competition master, created this dataset. It is part of a credible organization, Kaggle. The data was last refreshed two years ago in 2024, but the data I'm studying is for 2016, so it should still be relevant. Collection details are limited however.

● How are you addressing licensing, privacy, security, and accessibility?

-The dataset is released under CC0 Public Domain, so it is free to use, share and analyze without copyright concerns. The data source has also gotten the respondent's consent to use their data. If I was a part of that process, the respondents would be informed transparently about how their data would be used, and I would give them the right to inspect, update, or correct their data. The respondent's names and addresses are hashed into an ID so that their privacy is protected when the data is shown to the public and/or stakeholders. As far as security goes, the data is stored locally and version controlled. For accessibility, results will be shared using clear chart titles, high contrast colors, and plain language explanations.

● How did you verify the data's integrity?

-I made sure this data is primary, internal, structured data. I ensured that I downloaded it straight from Kaggle while it had a usability score of 9.41. I also saw that the suggestion tab had zero suggestions to improve the dataset even though it's been used by thousands of other analysts. I was also able to open the files fine in Excel.

● How does it help you answer your question?

-This data helps me answer the business task because it allows me to find trends from 3-12-16 to 5-12-16 across multiple fitness metrics to determine how and when people use the app. The data can tell us if the users are active, how active, if they're sedentary, if they sleep enough, how much energy they burn, and if they consistently track wellness. This can determine how Bellabeat markets the app to gain more subscribers and customers.

● Are there any problems with the data?

-Yes. Aside from the small sample size and the fact that this is old data, there are some other problems. I noticed that not every table had the same amount of ID's, so that means if I joined some tables together, there would be missing data in some columns. For instance, there are more users than sleep data (33 activity users, 24 sleep users). There is also missing demographics. There is no data about age, location, and income, and this is important because Bellabeat targets women. This is also device bias because users may differ from other smart device company users like Apple Watch users and Garmin users. There is also self-selection bias since users voluntarily submitted data; they may be more health-conscious than average consumers.

Key tasks

● Download data and store it appropriately.

-I downloaded the archive folder that contained both datasets that together span 3-12-16 to 5-12-16 and saved it to my Projects folder on my Windows PC.

● Identify how it's organized.

-The files are organized in two separate folders. Each folder has excel files containing one table each of a metric from the FitBit app of all respondents. Also, each folder's data spans two months.

- Sort and filter the data.

- I would sort total steps by ascending order and see how distance correlates to get an idea of whether someone was sedentary. I would filter to only see heart rates above 80 to see what percentage of people who use the smart device were active. I would sort the calories burned dates in ascending order to see what time of day most people work out with the smart device; same for intensities, steps, sleep. As for weight, I would sort by id first, then date second to get a table visual of how a respondent's weight changed chronologically.

- Determine the credibility of the data.

- Amazon Mechanical Turk is a legitimate crowdsourcing platform, but its credibility is highly debated. Since Google and Kaggle are vetting this dataset, I would say it is credible.

- Deliverable

- A description of all data sources used

- These files contain 30 survey respondent's personal tracker data, including minute-level

output for physical activity, heart rate, and sleep monitoring

Data Explorer

Version 2 (616.8 MB)

- ▼  mturkfitbit_export_3.12.16-4
 - ▼  Fitabase Data 3.12.16-4
 -  dailyActivity_merged.csv
 -  heartrate_seconds.csv
 -  hourlyCalories_merged.csv
 -  hourlyIntensities_merged.csv
 -  hourlySteps_merged.csv
 -  minuteCaloriesNarrow.csv
 -  minuteIntensitiesNarrow.csv
 -  minuteMETsNarrow.csv
 -  minuteSleep_merged.csv
 -  minuteStepsNarrow.csv
 -  weightLogInfo_merged.csv
- ▼  mturkfitbit_export_4.12.16-5
 - ▼  Fitabase Data 4.12.16-5
 -  dailyActivity_merged.csv
 -  dailyCalories_merged.csv
 -  dailyIntensities_merged.csv
 -  dailySteps_merged.csv
 -  heartrate_seconds.csv
 -  hourlyCalories_merged.csv
 -  hourlyIntensities_merged.csv
 -  hourlySteps_merged.csv
 -  minuteCaloriesNarrow.csv
 -  minuteCaloriesWide.csv
 -  minuteIntensitiesNarrow.csv
 -  minuteIntensitiesWide.csv
 -  minuteMETsNarrow.csv
 -  minuteSleep_merged.csv
 -  minuteStepsNarrow.csv
 -  minuteStepsWide.csv
 -  sleepDay_merged.csv
 -  weightLogInfo_merged.csv

Process

Case Study Roadmap - Process

Guiding questions

- What tools are you choosing and why?

-Because the data is over 1 million rows, I will be using Python for cleaning and analysts rather than Excel because it would be more efficient at handling the data. I will use the Pandas library for data manipulation, Matplotlib for data visualization, Jupyter Notebook for documentation and reproducibility, and I may use Tableau as well for data visualization.

● Have you ensured your data's integrity?

-I have replicated the data straight from the source, which does not continually update, so it should be accurate. The data only comes from one source though, so there may be insufficient data to come up with an accurate analysis. Here are integrity checks I will also perform on the data in Python: file import validation, column inspection, user counts, data type review, duplicate detection, and missing value review.

● What steps have you taken to ensure that your data is clean?

-First I will find any data errors. I will try to correct it, but if I can't and there is enough data for analysis, I may omit it so I can perform the analysis without the wrong data. If there is not enough data, I will ask stakeholders if there is time to collect more data or else proxy the data. Some data errors I will check for are null data and duplicate data. I will remove duplicate data. Then, I will check there is outdated, incorrect/inaccurate, or inconsistent data. I will convert date columns to the date datatype. After everything is done, I will merge activity and sleep data into a single analysis-ready dataset

● How can you verify that your data is clean and ready to analyze?

-I will verify the data is clean by rechecking the dataset, doing some manual clean ups if needed, and then taking a moment to sit back and think about the original purpose of the project. I will also look back on the changelog to make sure any changes I may not understand were explained and accounted for. I will use my data-cleaning verification checklist to recheck my data cleaning efforts. I will consider the business problem, then the goal, then I will consider whether the data is capable of solving the problem and meeting the project objectives.

● Have you documented your cleaning process so you can review and share those results?

-Yes. Documentation is maintained through Jupyter Notebook and comments in code.

Key tasks

● Check the data for errors.

● Choose your tools.

-Python

● Transform the data so you can work with it effectively.

● Document the cleaning process.

Deliverable

● Documentation of any cleaning or manipulation of data

-[bellabeat-data-cleaning-changelog.pdf](#)

Analyze

For this section, I will stick to the four phases of analysis: organize data, format and adjust data, get input from others, and transform data by observing relationships between data points and making calculations.

Case Study Roadmap – Analyze

Guiding questions

- How should you organize your data to perform analysis on it?

-To understand user wellness behavior, I need both dailyActivity_merged and sleepDay_merged in one place, so I merged them. This creates a dataset where each row contains user, date, steps, calories, activity, and sleep. This organization makes trend analysis easier.

- Has your data been properly formatted?

-Yes

- What surprises did you discover in the data?

-I will list out everything that surprised me: The first dataset did not have a sleepDay table. Sleep tracking participation was lower than activity tracking. Users spent over 16 hours sedentary. Most activity is light activity. Average steps are below 10k despite users owning fitness trackers.

- What trends or relationships did you find in the data?

-The strongest positive trends are: high sedentary behavior, moderate step-calorie relationship, many users sleep around 7 hours, sleep tracking is less common, most movement comes from light activity, and activity is least on Sundays.

- How will these insights help answer your business questions?

-These insights answer what trends exist and how they apply to Bellabeat customers, which influences Bellabeat's marketing strategy. Bellabeat Leaf can be marketed as a wellness coach, sleep companion, or habit building tool rather than an activity tracker.

Key tasks

- Aggregate your data so it's useful and accessible.

- Organize and format your data.

- Perform calculations.

- Identify trends and relationships.

Deliverable

- A summary of your analysis

-The Fitbit dataset revealed several important trends in smart device usage. Users averaged 7,638 steps per day, which falls the 10,000 step benchmark. Users also spent

approximately 16.5 hours per day sedentary, making inactivity one of the most significant findings in the analysis. Sleep data showed that users averaged approximately 7 hours of sleep per night, although sleep tracking participation was lower than activity tracking participation. Most physical activity consisted of light activity rather than vigorous exercise. A moderate positive relationship was observed between daily steps and calories burned. These findings suggest that Bellabeat has an opportunity to position the Leaf as a wellness companion that encourages movement, supports sleep improvement, and helps users build healthier daily habits. The Python code is available here [case-study-bellabeat-python](#)

Share

Case Study Roadmap - Share

Guiding questions

● Were you able to answer how smart device users use their devices and how these trends can apply to Bellabeat customers?

-Yes

● What story does your data tell?

-The story is that users not active in March. Activity picks up in April. Activity peaks in May.

- Smart device users are interested in tracking wellness, but their behaviors show room for improvement. Users are active enough to generate meaningful activity data, but they still spend a large amount of time sedentary. Their average step count is below the 10k step benchmark, and most activity is light rather than intense. Sleep data shows that users average about 7 hours of sleep, but fewer users track sleep compared with activity. Users are everyday wellness consumers who need gentle support, reminders, and habit-building tools.

● How do your findings relate to your original question?

-My findings help improve marketing strategy by showing Bellabeat can market movement reminds, promote achievable daily goals, should focus on wellness habits, can educate users about sleep insights, can promote recovery and sleep consistent, and can show progress through activity summaries

● Who is your audience? What is the best way to communicate with them?

-My primary audience is Bellabeat's executive and marketing team. This includes Urska Srsen and Sando Mur. The best way to communicate with them is by using a presentation with: a short business task summary, a simple explanation of the data source, clear visualizations, key findings in plain language, and marketing recommendations.

● Can data visualization help you share your findings?

-Yes.

- Is your presentation accessible to your audience?

-Yes. The axes are labeled. The points are a different color than the background, and the colors are colorblind friendly. The information has also been simplified. I've also thought about the audience to whom the viz will be shown and tried to frame it in the context of the business task so that the stakeholders have an easier time connecting to it.

Key tasks

- Determine the best way to share your findings.

- Create effective data visualizations.

- Present your findings.

-The presentation can be found here [bellabeat-presentation.pdf](#)

- Ensure your work is accessible.

Deliverable

- Supporting visualizations and key findings

-The viz and key findings can be found here [case-study-bellabeat-python](#)

Act

Case Study Roadmap - Act

Guiding questions

- What is your final conclusion based on your analysis?

-Smart device users are engaged in wellness tracking but struggle with consistent healthy habits, particularly around sedentary behavior and sleep.

- How could your team and business apply your insights?

-Assuming current marketing emphasizes “track your activity”, instead they could emphasize “build healthier daily habits”. Since a lot of users spend time sedentary, movement reminders could be implemented. Since sleep tracking was lower than activity tracking, sleep education could be implemented. A Bellabeat Membership could also leverage findings by providing: activity challenges, habit-building plans, and wellness coaching

- What next steps would you or your stakeholders take based on your findings?

-There are many things Bellabeat could do next. They could analyze actual Bellabeat user data. Fitbit data is not Bellabeat data. Questions this should answer are “how often do users open the Bellabeat app?” and “what features are used most?”. They should also segment customers because not users behave the same way. For instance, beginners have low activity and sleep improvers are more interested in recovery. This way, marketing can then be personalized. They should also do A/B testing on marketing campaigns, where campaign A is “tracking your activity” and campaign B is “build healthier daily habits” and

measure click-through rate and conversion rate. They should investigate why users stop tracking sleep, what features drive engagement, and whether sleep insights improve retention. They should also measure long-term behavior change by evaluating if users become more active over time, if sleep habits improve, and what features drive improvement.

- Is there additional data you could use to expand on your findings?

-Of course. I could analyze Bellabeat user data to answer most used features, session frequency, and engagement patterns. Customer demographics could age groups, income, geography, and lifestyle differences. Business sales data could answer best-selling products, seasonal trends, and marketing performance. Customer survey data could answer why customers purchase leaf, feature preferences. Lastly, CDC Sleep & Activity research could validate findings using a larger population.

Key tasks

- Create your portfolio.

- Add your case study.

- Practice presenting your case study to a friend or family member.

Deliverable

- Your top three recommendations based on your analysis

-My top three recommendations would be to position the Leaf as a wellness companion, create movement and habit-building campaigns, and promote sleep and recovery insights.